

KNN Regressor

KNN Regressor predicts a continuous value by averaging the target values of the k nearest data points.

output $\Rightarrow$

Decision Rule $\Rightarrow$

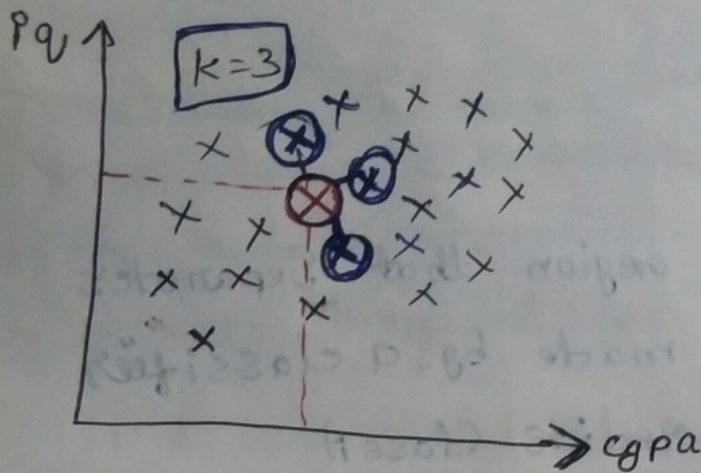
KNN Classifier

Class label

Majority vote

KNN Regressor

Continuous values
mean / weighted mean



cgpa | iq | lpa |

consider

3
5
4

$$\frac{12}{3} = 4 \text{ lpa}$$

k high $\rightarrow$ under fitting

k low $\rightarrow$ overfitting

Weighted KNN

weighted knn is a variation of knn where closer neighbors have more influence on the prediction than farther ones.
 → Near points matter more than far points

Problem with uniform knn:-

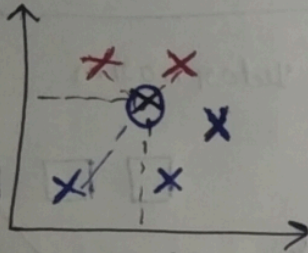
- all K neighbors contributes Equally
- a far neighbor can influence prediction as much as a close one. This can give wrong predictions

how weighted knn works:-

$$w_i = \frac{1}{d_i} \quad \text{or} \quad w_i = \frac{1}{d_i^2}$$

→ d_i ⇒ distance of i-th neighbor
 → Smaller distance → larger weights

Ex:-



i) uniform knn

red = 2
 blue = 3 } majority vote ⇒ Class blue

ii) weighted knn:-

		dist	wt	
1	→ red	→ 0.2	5	
2	→ red	→ 0.5	2	✓ red ⇒ 5 + 2 = 7
3	→ green	→ 1	1	
4	→ green	→ 2	0.5	
5	→ green	→ 3	0.33	green ⇒ 1 + 0.5 + 0.33

Hyperparameters in Knn Classifier:-

Knn has no training parameters, but its performance depends heavily on hyperparameters.

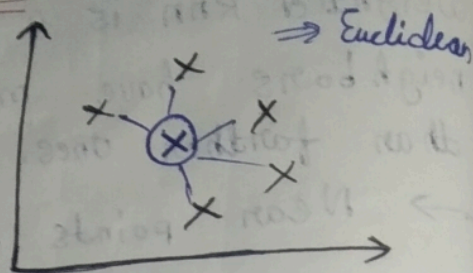
i) n-neighbors (k):- No. of nearest neighbors considered for prediction.

ii) weights:-

- 'uniform' → all neighbors equal
- 'distance' → closer neighbors have more influence.

metric (Distance Metric) :-

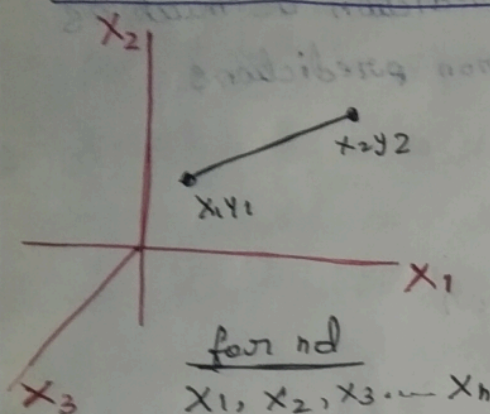
metric	meaning
Euclidean	→ straight line distance
Manhattan	city block distance
Minkowski	Generalized form



metric = 'minkowski'

$p = 1$ → Manhattan

$p = 2$ → Euclidean



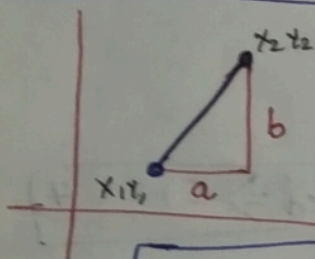
$$\text{dist} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

for nd $\Rightarrow \text{dist} = \sqrt{\sum_{i=1}^d (x_{2i} - x_{1i})^2}$

$$\text{dist} = \left(\sum_{i=1}^d (x_{2i} - x_{1i})^2 \right)^{1/2}$$

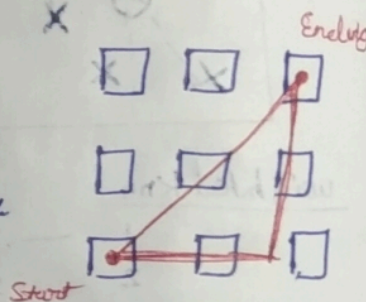
called as L2 norm in vector norm

manhattan distance



$a+b$ = manhattan distance

$$m = |x_2 - x_1| + |y_2 - y_1|$$



$$m = \sum_{i=1}^d |x_{2i} - x_{1i}|$$

Problem with euclidean

- should have same scale
Exp | Salary | $\Rightarrow$ here salary dominates the distance
- curse of dimensionality

minkowski :-

Euclidean = $\left(\sum_{i=1}^d (x_{2i} - x_{1i})^2 \right)^{1/2} \Rightarrow p=2 \Rightarrow \text{L2 norm}$

manhattan = $\left(\sum_{i=1}^d (|x_{2i} - x_{1i}|)^1 \right)^{1/1} \Rightarrow p=1 \Rightarrow \text{L1 norm}$

General $\Rightarrow \left(\sum_{i=1}^d (|x_{2i} - x_{1i}|)^p \right)^{1/p}$

$2 \leftarrow$ Eucl
 $1 \leftarrow$ manhattan

$$p > 0$$

Space and Time Complexity :-

Knn $\Rightarrow$ slow algorithm
 $\Downarrow$
 Prediction



during training
 will load data in
 a RAM

- $\rightarrow$ distance
- $\rightarrow$ store
- $\rightarrow$ K nearest
- $\rightarrow$ majority result

time Complexity of KNN = $O(nd)$

$\rightarrow n$ = no. of rows in training data / rows

$d \Rightarrow$ no. of features / col

$$\sqrt{\sum_{i=1}^d (x_{2i} - x_{1i})^2}$$

$x_1 \quad x_2 \quad x_3 \quad \dots \quad x_d \quad | \quad y$
 $\vdots$
 x_n

$(10^2, 1000)$

$\Rightarrow 10^8 \Rightarrow$ distances

Space Complexity $\Rightarrow O(nd)$

dataset size

algorithm = ('brute', 'auto', 'ball-tree', 'kd-tree')

The algorithm parameter decides how nearest neighbors are searched, Not how prediction is made.